12TH CHEMISTRY · CHAPTER 19 · SOLVED EXERCISE

Basics of Organic Chemistry

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. Multiple Choice Questions (MCQs):

I) Which of the following is organic in nature?

(a) CO_2
(b) CS_2
(c) HCN
(d) CCl_4

Answer: (d) CCl_4 — CCl_4 (tetrachloromethane) is a halogen derivative of methane. Oxides of carbon and cyanides are traditional inorganic compounds excluded from organic chemistry.

II) The empirical formula of $CH_3CH_2CH{=}CH_2$ matches with:

(a) CH_3CH_3
(b) $CH_2{=}CHCH_3$
(c) $CH_3CH_2CH_2CH_3$
(d) $CH{\equiv}CH$

Answer: (b) $CH_2{=}CHCH_3$ — $CH_3CH_2CH{=}CH_2$ is C_4H_8, empirical formula CH_2. Propene, C_3H_6, also has the empirical formula CH_2.

III) Compounds having single bonds usually undergo reactions:

(a) addition
(b) elimination
(c) substitution
(d) both elimination and substation

Answer: (c) substitution — Saturated compounds cannot add more atoms, so an atom or group is replaced by another, e.g., bromoethane with aqueous KOH.

IV) The functional group present in urea (H_2N–CO–NH_2) is:

(a) carbonyl
(b) amino
(c) amide
(d) all of these

Answer: (d) all of these — Urea has a C=O (carbonyl) group and two –NH_2 (amino) groups; together they make the amide group –$CONH_2$.

V) Which molecule is chiral and can rotate the plane polarised light?

(a) Pentan-1-ol
(b) Pentan-2-ol
(c) 2,2-dimethylpropan-1-ol
(d) 2-methylbutan-2-ol

Answer: (b) Pentan-2-ol — In pentan-2-ol, C-2 carries four different groups: CH_3, H, OH and $CH_2CH_2CH_3$. The other three have no chiral carbon.

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VI) Which carbon in $CH_3CH(CH_3)CH_2CH(OH)CH_3$ is chiral?

(a) C–1 (b) C–2

(c) C–3 (d) C–4

Answer: (d) C–4 — The carbon bearing the –OH group carries four different groups: H, OH, CH_3 and $CH_2CH(CH_3)_2$. Counting from the left of the formula as written, it is C-4.

VII) In what ways could two compounds of molecular formula $C_2H_2Br_2$ be related to each other?

(a) optical isomers (b) cis-trans isomers

(c) constitutional isomers (d) both b and c

Answer: (d) both b and c — 1,1-dibromoethene and 1,2-dibromoethene are constitutional isomers; cis- and trans-1,2-dibromoethene are cis-trans isomers. None is chiral.

VIII) Which of following shows cis-trans isomerism?

(a) 1-butene (b) propene

(c) 2-butene (d) 1,1-dichloro ethene

Answer: (c) 2-butene — In 2-butene each doubly bonded carbon carries two different groups (H and CH_3). In the others one doubly bonded carbon carries two identical groups.

IX) Which one of the following pairs do the isomers have identical boiling points?

(a) cis and trans isomers (b) (+) and (–) enantiomers

(c) cis-trans and optical isomers (d) structural isomers

Answer: (b) (+) and (–) enantiomers — Enantiomers have identical physical and chemical properties except the direction in which they rotate plane polarised light.

MCQ VI: by IUPAC numbering the compound is 4-methylpentan-2-ol and the same chiral carbon is numbered C-2; the answer (d) follows the formula as written from left to right. MCQ III: "substation" is printed for "substitution". The exercise has nine MCQs.

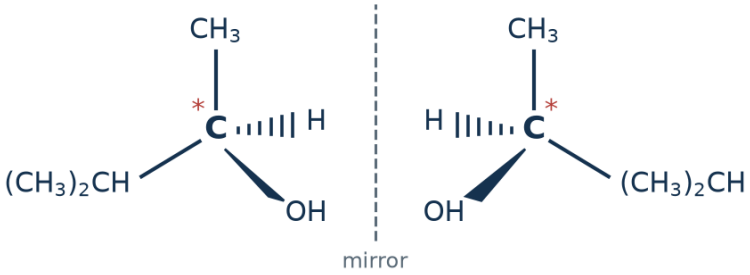
Q2. SHORT ANSWER QUESTIONS

a) Define enantioers. Give enantiomers of 2-methylbutan-2-ol.

Enantiomers are two stereoisomers which are non-superimposable mirror images of each other. They have identical physical and chemical properties but rotate the plane of polarised light in opposite directions to the same extent.

2-Methylbutan-2-ol, CH_3–C(OH)(CH_3)–CH_2–CH_3, has **no chiral carbon**: C-2 carries two identical CH_3 groups. Its mirror image is superimposable on it, so it is achiral and has no enantiomers.

The compound named in the question cannot have enantiomers. Its position isomer 3-methylbutan-2-ol, CH_3–CH(OH)–$CH(CH_3)_2$, has a chiral C-2; its enantiomers are shown below.



Enantiomers of 3-methylbutan-2-ol

b) Alkanes have no functional group. Justify.

The type of an organic compound is dictated by two factors in its functional group: (i) the polarity of the bond and (ii) the bond order, i.e., the number of bonds. Alkanes (and cycloalkanes) contain only non-polar C–C and C–H single bonds. They possess no charges or polarity and no double or triple bonds in their structures, so they have no functional group.

c) Deduce the empirical and molecular formulae of the following compounds.

	Structure given	Molecular formula	Empirical formula
a)	Cyclohexane (six-membered ring, all single bonds)	C_6H_{12}	CH_2
b)	Open chain of five carbon atoms with two double bonds	C_5H_8	C_5H_8
c)	Four-membered ring with two double bonds	C_4H_4	CH

Each corner or line end is one carbon atom; hydrogen atoms are added to give every carbon four bonds.

d) Give systematic names of the following compounds.

	Formula	Systematic name
i)	$CH_3–CH_2–CN$	propanenitrile
ii)	$CH_3COCH_2CH_2–CH_3$	pentan-2-one
iii)	$CH_3CH_2COOCH_3$	methyl propanoate

e) Define electrophile and nucleophile with two examples each.

Electrophile: an atom or group of atoms that is able to accept a pair of electrons to form a new bond. Examples: H^+, NO_2^+ (also CH_3^+, Br^+, SO_3, BF_3).

Nucleophile: an atom or group of atoms that is able to donate a pair of electrons to form a new bond. Examples: OH^-, NH_3 (also CN^-, Br^-, H_2O).

f) Pent-2-ene shows cis-trans isomerism but pent-1-ene does not. Give reason.

In pent-2-ene, CH_3–CH=CH–CH_2–CH_3, each doubly bonded carbon carries two different groups (H and CH_3 on C-2; H and C_2H_5 on C-3), so two arrangements, cis and trans, are possible. In pent-1-ene, CH_2=CH–CH_2–CH_2–CH_3, C-1 is bonded to two identical hydrogen atoms, so only one arrangement exists and cis-trans isomerism is not possible.

g) Define: i) Constitutional isomerism ii) Stereoisomerism

i) Constitutional (structural) isomerism: isomerism in which the same atoms or groups are connected at different places in two or more molecules, i.e., the isomers have different bond connectivity.

ii) Stereoisomerism: isomerism in which the isomers have the same bond connectivity, but their substituents or groups have a different spatial arrangement (different configuration).

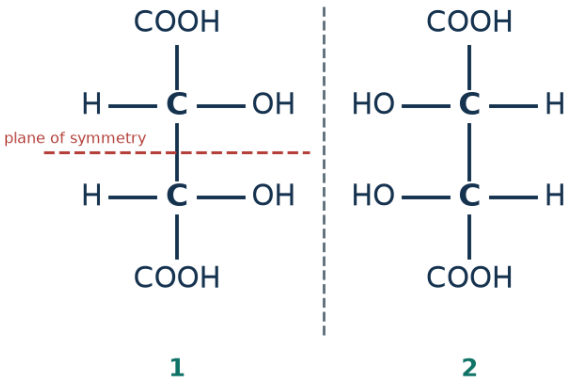
h) Define: i) racemic mixture ii) Optical activity

i) Racemic mixture: an optically inactive mixture containing 50% dextrorotatory and 50% levorotatory enantiomers. It is shown by the symbol (±).

ii) Optical activity: the property of a compound of rotating the plane of plane polarised light.

i Explain why meso tartaric acid is optically inactive although it has two chiral carbon atoms.

Meso tartaric acid has two similar chiral carbon atoms and a plane of symmetry, which divides the molecule into two halves that are mirror images of each other. The two halves rotate the plane of polarised light in opposite directions to the same extent. The rotation of one half is internally compensated (cancelled) by the other, so the molecule is optically inactive.



Meso tartaric acid

Q3. CONSTRUCTED RESPONSE QUESTION

a) A compound $C_6H_5CH_3$ reacts with Cl_2 in the presence of sunlight to form the product $C_6H_5CCl_3$. Name the mechanism of the reaction. Give its different steps.

The mechanism is **free radical substitution**: the three hydrogen atoms of the methyl group are replaced one by one by chlorine atoms.

$$C_6H_5CH_3 + 3Cl_2 \rightarrow C_6H_5CCl_3 + 3HCl \text{ (sunlight)}$$

1. Initiation: sunlight breaks the Cl–Cl bond homolytically, giving chlorine free radicals.

$$Cl\text{–}Cl \rightarrow Cl\bullet + Cl\bullet$$

2. Propagation: the radicals generate more radicals and the product.

$$C_6H_5CH_3 + Cl\bullet \rightarrow C_6H_5CH_2\bullet + HCl$$
$$C_6H_5CH_2\bullet + Cl_2 \rightarrow C_6H_5CH_2Cl + Cl\bullet$$

The same two steps are repeated with $C_6H_5CH_2Cl$ and then with $C_6H_5CHCl_2$, until $C_6H_5CCl_3$ is formed.

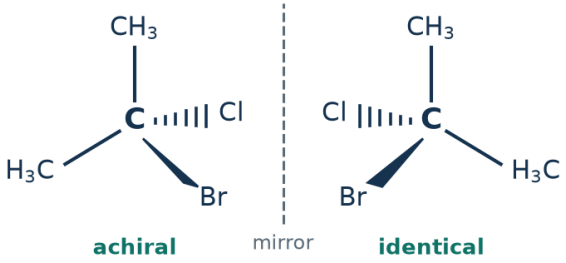
3. Termination: free radicals of the same or different types combine.

$$Cl\bullet + Cl\bullet \rightarrow Cl_2$$
$$C_6H_5CH_2\bullet + Cl\bullet \rightarrow C_6H_5CH_2Cl$$
$$C_6H_5CH_2\bullet + \bullet CH_2C_6H_5 \rightarrow C_6H_5CH_2\text{–}CH_2C_6H_5$$

b) Draw the mirror image of each of the following compounds. Label each molecule as chiral or achiral.

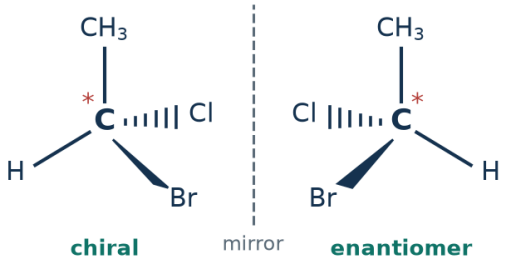
i) 2-bromo-2-chloropropane

CH_3–$CBrCl$–CH_3: the central carbon carries two identical CH_3 groups. The mirror image is superimposable on the molecule: **achiral**.



ii) 1-bromo-1-chloroethane

CH_3–$CHBrCl$: C-1 carries four different groups (H, CH_3, Br, Cl). The mirror image is non-superimposable: **chiral**.

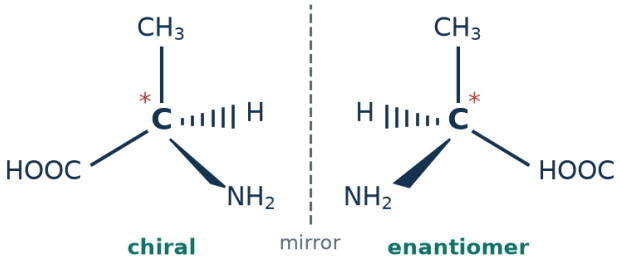


iii) ethoxyethane

CH_3–CH_2–O–CH_2–CH_3 has no carbon with four different groups. Its mirror image, CH_3–CH_2–O–CH_2–CH_3, is identical with it: **achiral**.

iv) 2-aminopropanoic acid

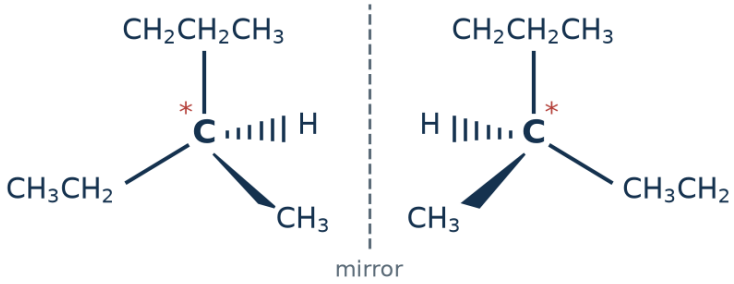
CH_3–CH(NH_2)–COOH: C-2 carries four different groups (H, CH_3, NH_2, COOH). The mirror image is non-superimposable: **chiral**.



c) What is the alkane of lowest relative molecular mass that contains a chiral centre? Give its systematic name and molecular formula. Also draw mirror image of this compound.

A chiral carbon in an alkane must carry four different groups. The smallest possible set is H, CH_3, C_2H_5 and C_3H_7, which gives seven carbon atoms. **Systematic name:** 3-methylhexane, CH_3–CH_2–CH(CH_3)–CH_2–CH_2–CH_3. **Molecular formula:** C_7H_{16} (relative molecular mass 100).

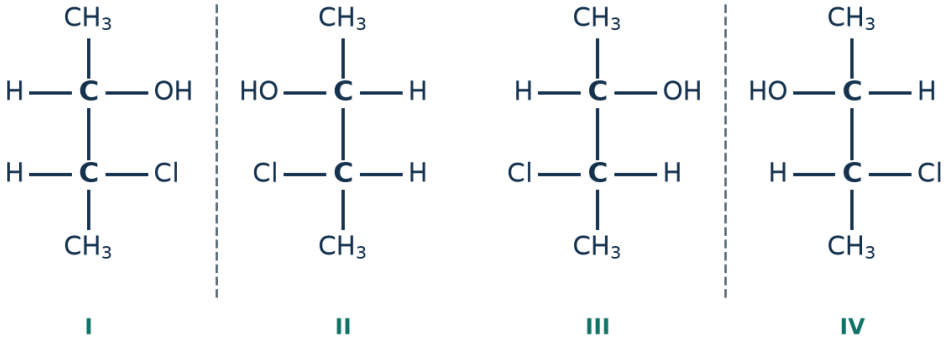
2,3-Dimethylpentane, also C_7H_{16}, is chiral as well; either is accepted as the lowest-mass chiral alkane.



3-Methylhexane and its mirror image

d) 3-chlorobutan-2-ol is an optically active compound. Draw all its possible optical isomers. Is there any meso form of this compound? Give reason.

CH_3–CH(OH)–CH(Cl)–CH_3 has two chiral carbon atoms (C-2 and C-3), so the number of optical isomers is 2^2 = **4**. I and II are one pair of enantiomers; III and IV are the other pair.



The four optical isomers of 3-chlorobutan-2-ol

No meso form: a meso form needs two *similar* chiral carbon atoms and a plane of symmetry. Here the two chiral carbons are different (one carries –OH, the other –Cl), so no isomer has a plane of symmetry and all four are optically active.

DESCRIPTIVE QUESTIONS

Q4. Explain the difference between homolytic and heterolytic fissions of a covalent bond.

Homolytic fission (homolysis) occurs when the breaking bond is non-polar. The two shared electrons of the bond are distributed **equally** between the two atoms; each atom gets one electron. The atoms or groups of atoms with unpaired electrons that are formed are called **free radicals**. Free radicals are reaction intermediates and are highly reactive because of the tendency of their unpaired electrons to be paired.

$$Cl–Cl \rightarrow Cl\bullet + Cl\bullet$$

$$H_3C–H \rightarrow \bullet CH_3 + H\bullet$$

Heterolytic fission (heterolysis) occurs when the breaking bond is polar. The two shared electrons are distributed **unequally**: one atom keeps both electrons and acquires a negative charge; the other atom is deficient in one electron and acquires a positive charge. Ions are formed. A species containing a carbon atom with a positive charge is a **carbocation**; it forms when carbon is bonded to a heteroatom more electronegative than carbon (O, N, S, P, halogens). A **carbanion** forms when carbon is bonded to a heteroatom (metal) less electronegative than carbon (Li, Mg, Cd).

$$(CH_3)_3C–Cl \rightarrow (CH_3)_3C^+ + Cl^-$$

Homolytic fission	Heterolytic fission
The breaking bond is non-polar.	The breaking bond is polar.
The shared electrons are divided equally: one to each atom.	The shared electrons are divided unequally: both go to one atom.
Free radicals (neutral, with an unpaired electron) are formed.	Ions are formed: a cation (e.g., a carbocation) and an anion.
Cl–Cl → Cl• + Cl•	$(CH_3)_3C–Cl \rightarrow (CH_3)_3C^+ + Cl^-$
Leads to free radical mechanisms.	Leads to ionic or polar mechanisms.

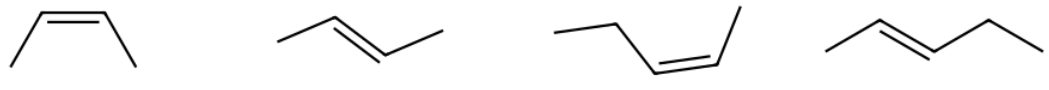
Q5. Discuss cis-trans isomerism. Explain with two examples.

Stereoisomers having a different spatial arrangement of atoms due to **restricted rotation about a double bond** in open chain compounds are called **cis-trans isomers**.

Free rotation about a single bond: in butane the C–C single bond between C-2 and C-3 rotates freely, so the methyl groups also rotate and form I changes into form II and back. Forms I and II are **conformational isomers**: they are interconvertible by free rotation and cannot be separated at room temperature.

Restricted rotation about a double bond: in but-2-ene, $CH_3–CH=CH–CH_3$, carbon atoms 2 and 3 are sp^2 hybridized. The C=C double bond consists of a σ bond (linear overlap of sp^2 hybrid orbitals) and a π bond (parallel overlap of p orbitals). The π bond locks the molecule in one position: C-2, C-3 and all the groups attached to them lie in the same plane and their positions in space are fixed. For free rotation the π bond must first be broken, which requires 263 kJ/mol; this is not possible at room temperature. This restriction of rotation is responsible for cis-trans isomerism.

The **cis** isomer has the two similar groups on the same side of the double bond; the **trans** isomer has them on opposite sides.



cis-but-2-ene trans-but-2-ene cis-pent-2-ene trans-pent-2-ene

Example 1, but-2-ene: in cis-but-2-ene the two CH_3 groups are on the same side of the double bond; in trans-but-2-ene they are on opposite sides. **Example 2, pent-2-ene:** the CH_3 and C_2H_5 groups are on the same side in the cis isomer and on opposite sides in the trans isomer.

Condition: for an open chain compound the necessary and sufficient condition is that it has a double bond and that two different groups are attached to each of the doubly bonded carbon atoms. Cis-trans isomerism is not possible if one of the unsaturated carbon atoms is bonded to two identical groups, e.g., but-1-ene ($CH_2=CH–CH_2–CH_3$) and 2,3-dimethylbut-2-ene, $(CH_3)_2C=C(CH_3)_2$.

Cis-trans isomers have different physical and chemical properties. Trans isomers are more stable than the corresponding cis isomers and hence differ in chemical reactivity. They also differ in physical properties such as melting point, boiling point, dipole moment, density and heat of vapourisation.

Q6. Describe the phenomenon of optical activity. Explain with an example.

Optical isomers are stereoisomers which have identical physical and chemical properties except their ability to rotate plane polarised light. Such isomers are said to be optically active, and this property of rotating the plane of polarisation is called **optical activity**. It is identified with plane polarised light (PPL). If the compound rotates the plane of polarised light to the right (clockwise) it is **dextrorotatory (+)**; if to the left (anticlockwise) it is **levorotatory (–)**.

A **polarimeter** is used to identify and measure quantitatively the optical activity of a compound using PPL. PPL is monochromatic light of a single wavelength which moves in only one plane. A solution of the compound is placed before the PPL: if it rotates the PPL it is optically active, and if not, optically inactive.

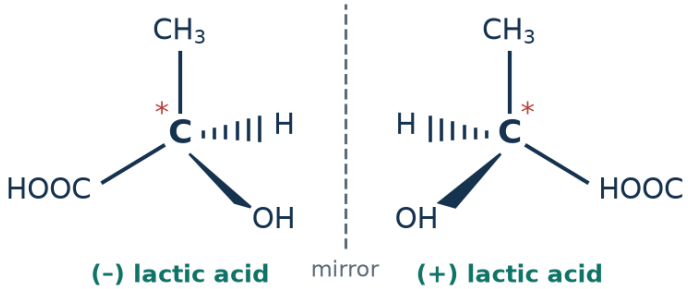
Unpolarised light → polarizer → PPL → sample → plane rotated clockwise (dextrorotatory) or anticlockwise (levorotatory).

The basic requirement for a compound to be optically active is that it is **non-superimposable on its mirror image**. This condition is fulfilled by asymmetric (chiral) molecules, just as our right and left hands are non-superimposable. The word “chiral” is from the Greek *cheir*, hand; chirality is also called handedness. Molecules that are superimposable on their mirror images are **achiral** and optically inactive.

A carbon atom bearing four different atoms or groups of atoms is called a **chiral carbon** (chiral centre). A molecule with only one chiral carbon is always asymmetric and hence optically active. If there is more than one chiral carbon, the molecule may or may not be asymmetric and optically active.

Example:

Lactic acid, CH_3–CH(OH)–COOH, has one chiral carbon bonded to four different groups: CH_3, H, OH and COOH. Its three-dimensional structure has two stereoisomers which are non-superimposable mirror images of each other, called **enantiomers** (“enantio” means opposite). Enantiomers have identical physical and chemical properties, but they rotate the plane of polarised light in opposite directions to the same extent: one is dextrorotatory (+), the other levorotatory (–).



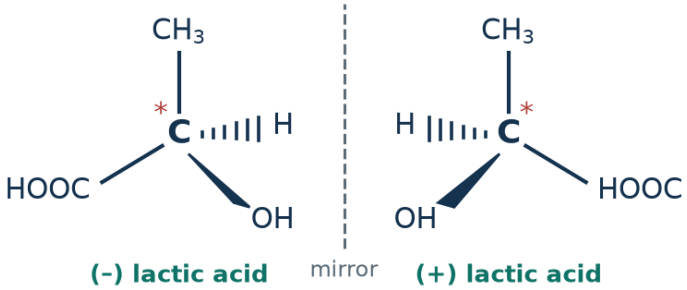
Enantiomers of lactic acid (= chiral carbon)*

When 50% dextrorotatory and 50% levorotatory enantiomers are present, they form an optically inactive mixture called a **racemic mixture**. It is symbolised by (±) in front of the name. Thus lactic acid exists in two optically active forms, (+) lactic acid and (–) lactic acid, and an inactive racemic mixture (±).

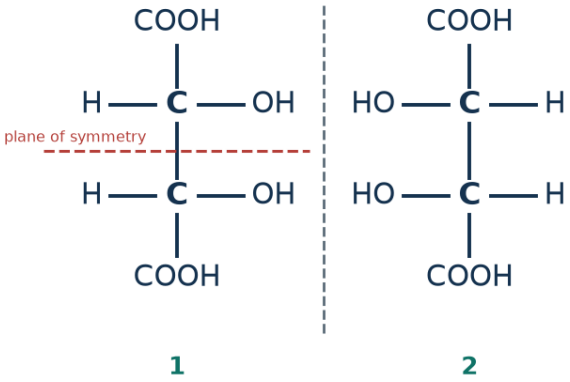
Q7. What is the difference between racemic mixture and meso compound? Draw the structures of enantiomers and meso form of lactic acid.

Racemic mixture	Meso compound
A mixture of two compounds: 50% (+) and 50% (–) enantiomers.	A single compound.
Each molecule is chiral and optically active.	The molecule is achiral: it has similar chiral carbons and a plane of symmetry.
Optically inactive by external compensation: the rotations of the two enantiomers cancel.	Optically inactive by internal compensation: the rotations of the two halves of the molecule cancel.
Can be separated into two optically active enantiomers.	Cannot be separated into optically active forms.
Shown by (±), e.g., (±) lactic acid.	e.g., meso tartaric acid.

Enantiomers of lactic acid:



Meso form: lactic acid has only one chiral carbon atom, so it has no meso form; it exists as (+) lactic acid, (–) lactic acid and the racemic mixture (±). The textbook's example of a meso form is tartaric acid, which has two similar chiral carbons:



Meso tartaric acid

The question asks for the "meso form of lactic acid", which does not exist; the meso form of tartaric acid is drawn instead. The terms external and internal compensation and the point about separation are standard; the textbook says "internally compensated" for the meso form.

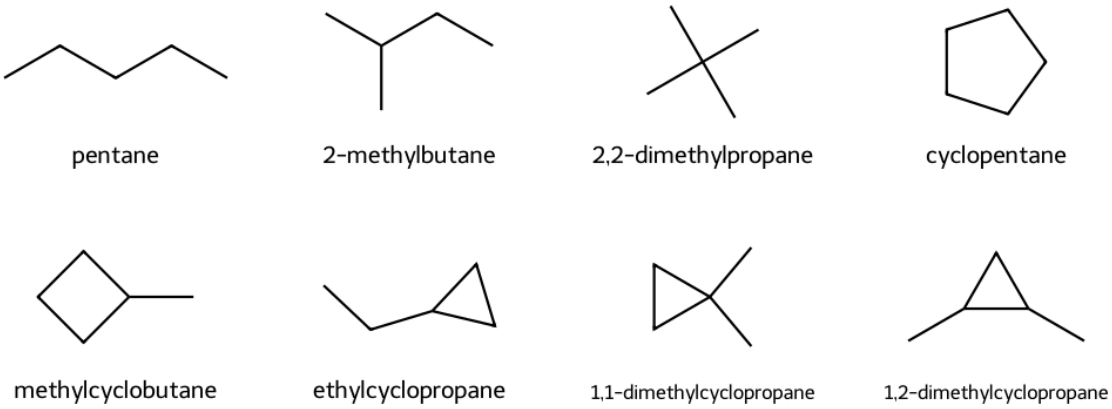
QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 19.1

a) Draw the different possible structures containing five carbon atoms.

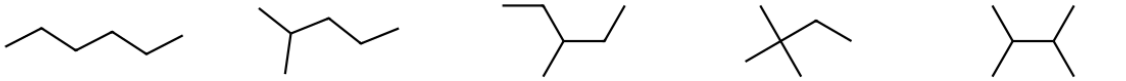
Five carbon atoms can be joined as three open chains and five rings (carbon skeletons shown as bond-line structures):



Open chains: pentane, 2-methylbutane, 2,2-dimethylpropane. Rings: cyclopentane, methylcyclobutane, ethylcyclopropane, 1,1- and 1,2-dimethylcyclopropane.

b) Give different open chain structures containing six carbon atoms.

There are five open chain arrangements of six carbon atoms:



hexane 2-methylpentane 3-methylpentane 2,2-dimethylbutane 2,3-dimethylbutane

They are hexane, 2-methylpentane, 3-methylpentane, 2,2-dimethylbutane and 2,3-dimethylbutane.

Quick Check 19.2

a) Classify the following compounds as an aliphatic, alicyclic or aromatic.

No.	Structure given	Class
i)	Cyclohexane ring carrying a CH_3 group (methylcyclohexane)	Alicyclic
ii)	Six-membered ring with two double bonds carrying a CH_3 group	Alicyclic
iii)	Benzene ring carrying a CH_3 group (methylbenzene)	Aromatic
iv)	Two fused benzene rings (naphthalene)	Aromatic
vii)	Cyclohexane	Alicyclic

No structures are printed for parts v) and vi) in the textbook, so they cannot be classified. None of the printed structures is aliphatic (open chain).

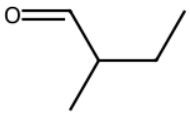
Quick Check 19.3

a) What are the systematic names of the following?

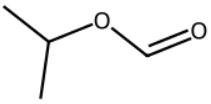
No.	Formula	Systematic name
i)	$(CH_3)_3COH$	2-methylpropan-2-ol
ii)	CH_3CCl_3	1,1,1-trichloroethane
iii)	$CH_2=CH–CH_2–OH$	prop-2-en-1-ol
iv)	$CH_3COC_2H_5$	butan-2-one (butanone)
v)	$CH_2=C(CH_3)COOCH_3$	methyl 2-methylprop-2-enoate

b) Draw the structures of the following.

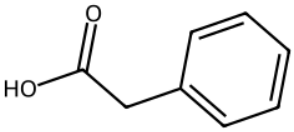
No.	Name	Condensed structure
i)	2-methylbutanal	CH_3–CH_2–CH(CH_3)–CHO
ii)	prop-2-ylmethanoate	HCOO–CH(CH_3)$_2$
iii)	phenylethanoic acid	C_6H_5–CH_2–COOH
iv)	cyclohexanone	six-membered carbon ring with one C=O; $C_6H_{10}O$
v)	3-methylpentanal	CH_3–CH_2–CH(CH_3)–CH_2–CHO



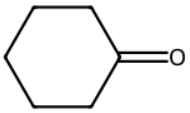
i) 2-methylbutanal



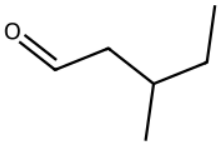
ii) prop-2-yl methanoate



iii) phenylethanoic acid



iv) cyclohexanone



v) 3-methylpentanal

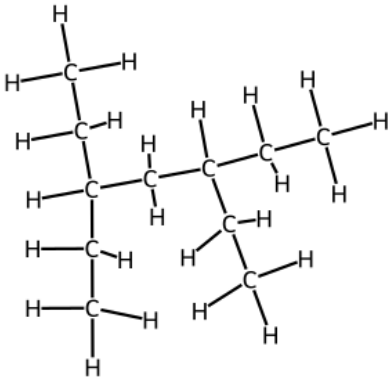
The skeletal structures above show the same five compounds.

Quick Check 19.4

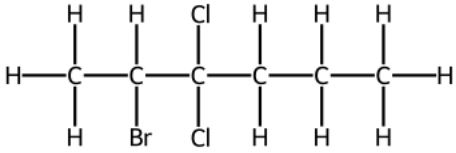
a) Draw the displayed and skeletal formulae of the following: i. 3,5-diethylheptane ii. 2-bromo-3,3-dichlorohexane iii. 2-methylbutanal

Condensed formulae: i. $CH_3CH_2CH(C_2H_5)CH_2CH(C_2H_5)CH_2CH_3$; ii. $CH_3CH(Br)CCl_2CH_2CH_2CH_3$; iii. $CH_3CH_2CH(CH_3)CHO$.

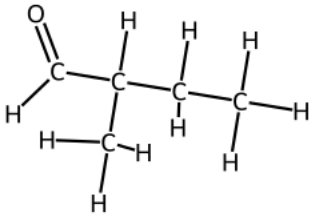
Displayed formulae (all atoms and bonds shown):



i. 3,5-diethylheptane

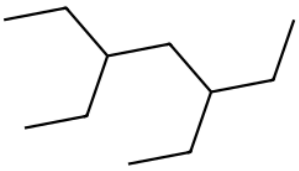


ii. 2-bromo-3,3-dichlorohexane

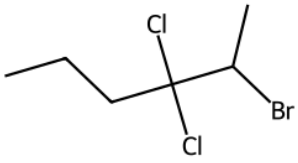


iii. 2-methylbutanal

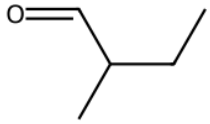
Skeletal formulae:



i. 3,5-diethylheptane



ii. 2-bromo-3,3-dichlorohexane

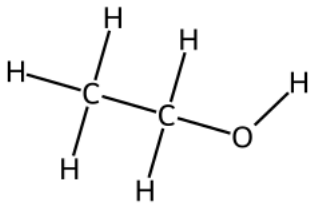


iii. 2-methylbutanal

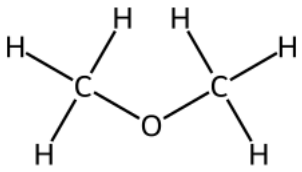
In the skeletal formulae each corner or line end is a carbon atom; hydrogen atoms on carbon are not drawn.

b) Draw the displayed and condensed structures of ethanol and methoxy methane, and explain whether they are isomers of the same molecule or not.

Displayed formulae:



ethanol

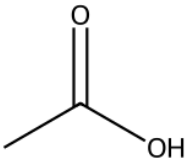


methoxymethane

Condensed formulae: ethanol CH_3CH_2OH; methoxymethane CH_3OCH_3. Both have the same molecular formula, C_2H_6O, but different structures and different functional groups (alcohol –OH and ether C–O–C). So they are isomers of each other (functional group isomers), not the same compound.

c) Draw the skeletal formula of ethanoic acid (CH_3COOH).

The line end is the CH_3 carbon and the corner is the carboxyl carbon; the C=O and O–H groups are shown.



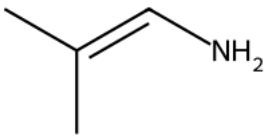
Quick Check 19.5

a) Write molecular formula for the compound represented by the following bond-line structure:

The structure is Br–CH_2–CH_2–COOH (3-bromopropanoic acid): 3 C atoms, 5 H atoms, 1 Br atom and 2 O atoms. Molecular formula: **$C_3H_5BrO_2$**.

b) Draw the skeletal formula for $(CH_3)_2C=CHNH_2$

The two line ends are the CH_3 groups; the C=C double bond and the NH_2 group are shown.



c) Write molecular formula for the compound represented by the following bond-line structure:

The structure is a cyclohexane ring in which one ring carbon carries a methyl group and a –CH=CH–CH_3 group. Ring: 6 C and 10 H; methyl: CH_3; side chain: C_3H_5. Molecular formula: **$C_{10}H_{18}$**.

d) Give the molecular formulae of the following.

i) 1,2-dichlorocyclohexane: **$C_6H_{10}Cl_2$**. ii) Benzene ring carrying $-CH_2Cl$ and $-CO_2H$ groups on adjacent carbons: **$C_8H_7ClO_2$**.

In part ii) the side group is printed as "CH_3Cl"; a group attached to the ring must be $-CH_2Cl$.

Quick Check 19.6

a) Describe each of the following as electrophile, nucleophile or free radical.

No.	Species	Type	Reason
i)	NH_3	Nucleophile	Lone pair on nitrogen can be donated
ii)	Br	Free radical	A bromine atom has an unpaired electron
iii)	H_2O	Nucleophile	Lone pairs on oxygen
iv)	NO_2^+	Electrophile	Positive ion; accepts an electron pair
v)	$Cl^{\delta+}$	Electrophile	Electron-deficient (partially positive) chlorine

b) Classify each of these reactions as addition, substitution, elimination or rearrangement.

No.	Reaction	Type
(I)	$CH_3CH{=}CH_2 + HBr \rightarrow CH_3CHBrCH_3$	Addition
(II)	$CH_4 + Br_2 \rightarrow CH_3Br + HBr$	Substitution
(III)	$CH_3CH_2CHClCH_3 \rightarrow CH_3CH_2CH{=}CH_2 + HCl$	Elimination

c) Classify the following reactions into one (or more) of the following six categories: Substitution, addition, hydrolysis, reduction, condensation, oxidation.

No.	Reaction	Category
(i)	$(CH_3)_2C{=}O + H_2N{-}NH_2 \rightarrow (CH_3)_2C{=}N{-}NH_2 + H_2O$	Condensation (two molecules join with loss of water)
(ii)	$CH_3CHO + 2[H] \rightarrow CH_3CH_2OH$	Reduction (gain of hydrogen); it is also an addition

Quick Check 19.7

Q1: Classify the following objects as chiral or achiral. (a) Pencil (b) Book (c) Shoe (d) Cup (e) Ear (f) Foot (g) Egg (h) Spoon

Chiral (not superimposable on the mirror image): shoe, ear, foot. **Achiral** (have a plane of symmetry): pencil, cup, egg, spoon and a plain book.

A book is achiral only as a plain object; if its printing and the side on which it opens are considered, it is chiral.

Q2: Glucose and fructose are two monosaccharides. i. How many chiral carbon atoms are there in glucose and fructose? ii. Calculate the possible number of optical isomers of glucose and fructose. iii. Is there any meso form in glucose or fructose? Give reason.

i. In the open chain forms, glucose, $CH_2OH–(CHOH)_4–CHO$, has **4** chiral carbon atoms (C-2 to C-5); fructose, $CH_2OH–CO–(CHOH)_3–CH_2OH$, has **3** (C-3 to C-5).

ii. Number of optical isomers = 2^n. Glucose: 2^4 = **16**. Fructose: 2^3 = **8**.

iii. No. A meso form needs similar chiral carbons and a plane of symmetry. In glucose the two ends are different (–CHO and $–CH_2OH$), and in fructose the C=O group is at C-2, so neither molecule can be divided into two mirror-image halves.

The open chain structures of glucose and fructose are not given in this chapter; they are used here from general knowledge.

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